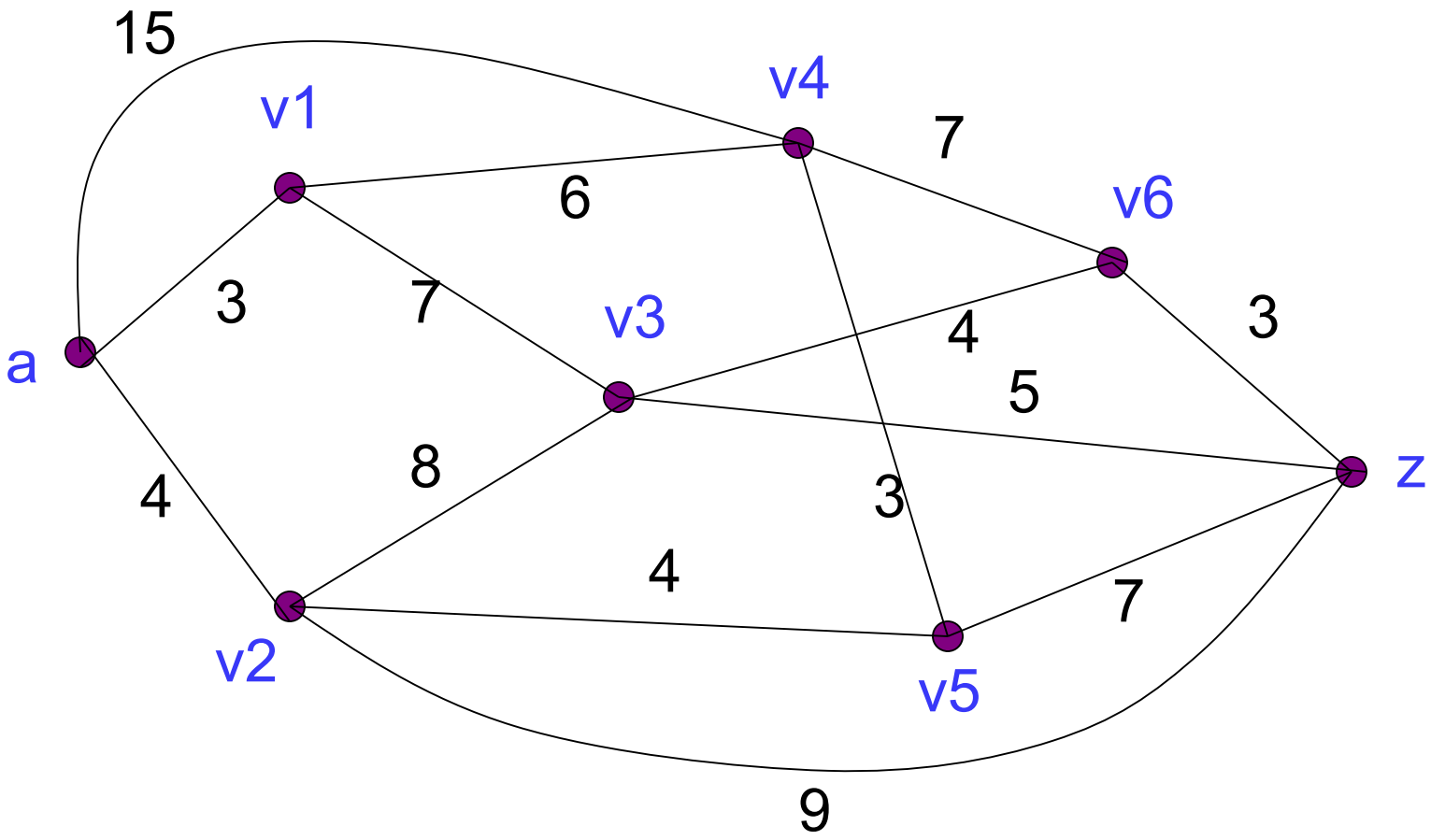
Shortest Path Problem

Shortest Path Problems

- Let *G* be a weighted graph.
- Let *u* and *v* be two vertices in *G*, and let *P* be a path in *G* from *u* to *v*.
- The length of path *P*, written *L*(*P*), is the sum of the weights of all the edges on path *P*.
- A shortest path from a vertex to another vertex is a path with the shortest length between the vertices.

example



Dijkstra's Shortest Path Algorithm

1. $S := \emptyset$
2. $N := V$
3. For all vertices, $u \in V$, $u \neq a$, $L(u) := \infty$
4. $L(a) := 0$

Dijkstra's Shortest Path Algorithm

5. While $z \notin S$ do,

5.a Let $v \in N$ be such that

$L(v)=\min\{L(u) \mid u \in N\}$

5.b $S := S \cup \{v\}$

5.c $N := N - \{v\}$

Dijkstra's Shortest Path Algorithm

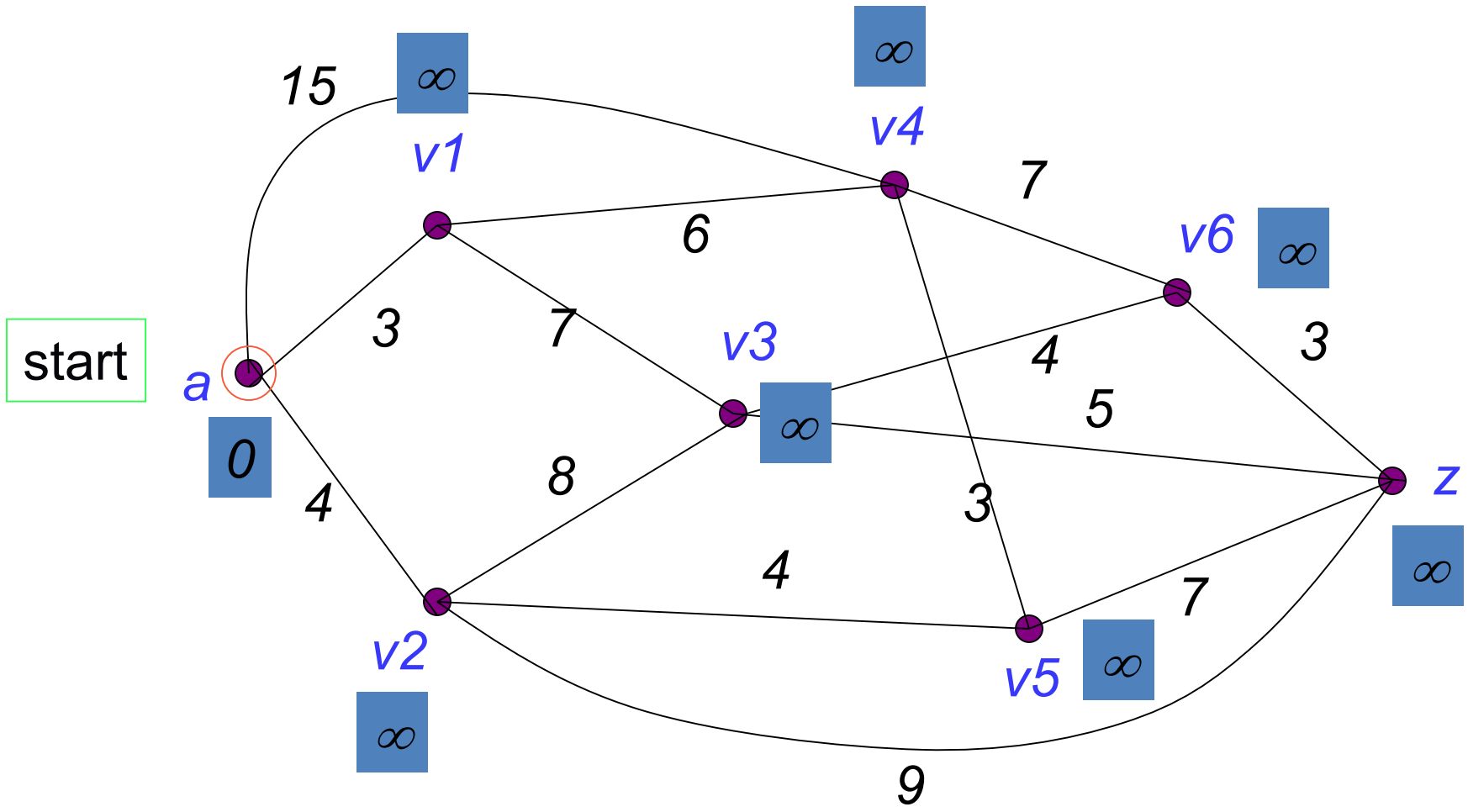
5.d For all $w \in N$ such that there is an edge from v to w

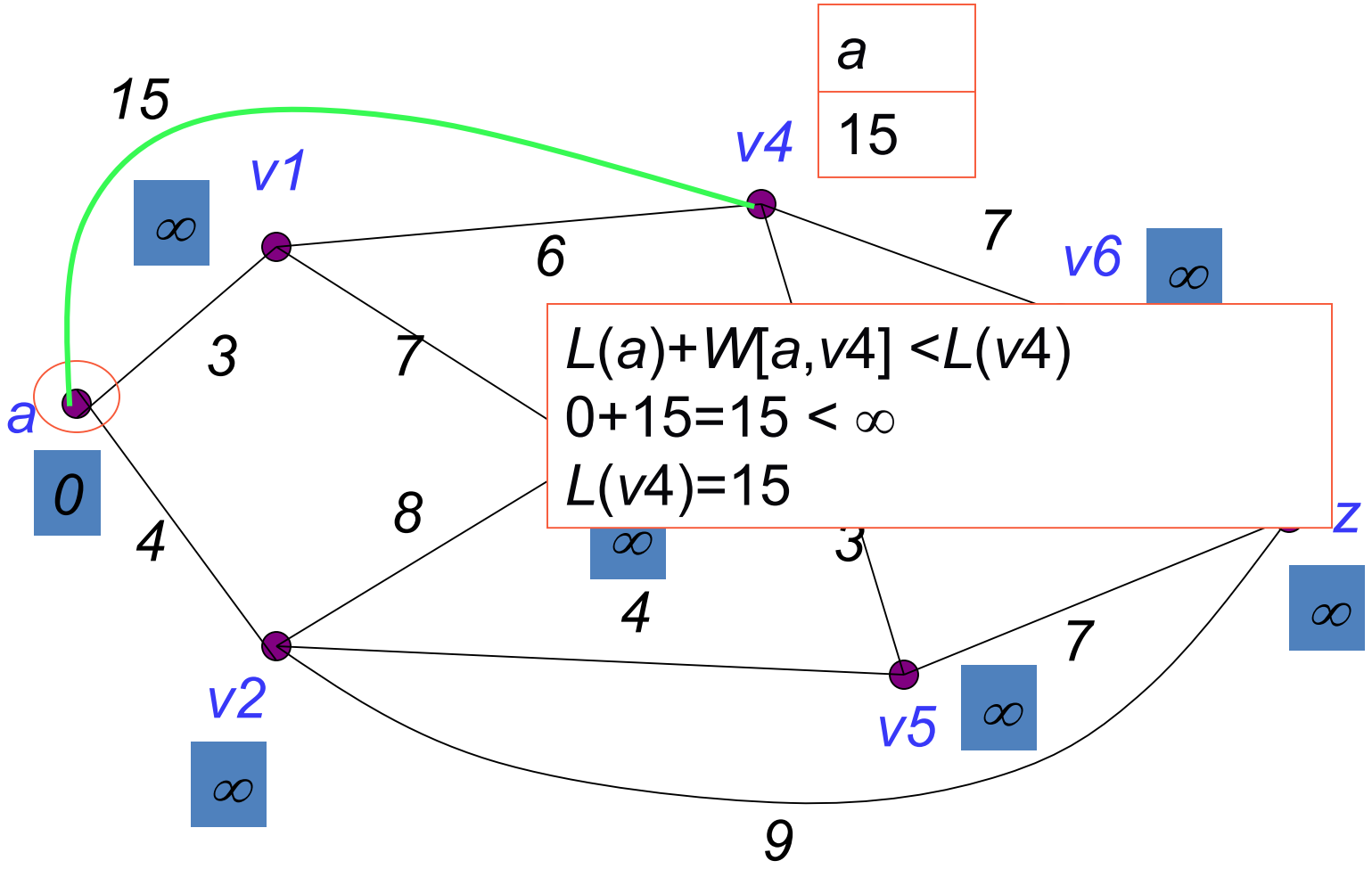
5.d.1 if $L(v)+W[v,w] < L(w)$ then

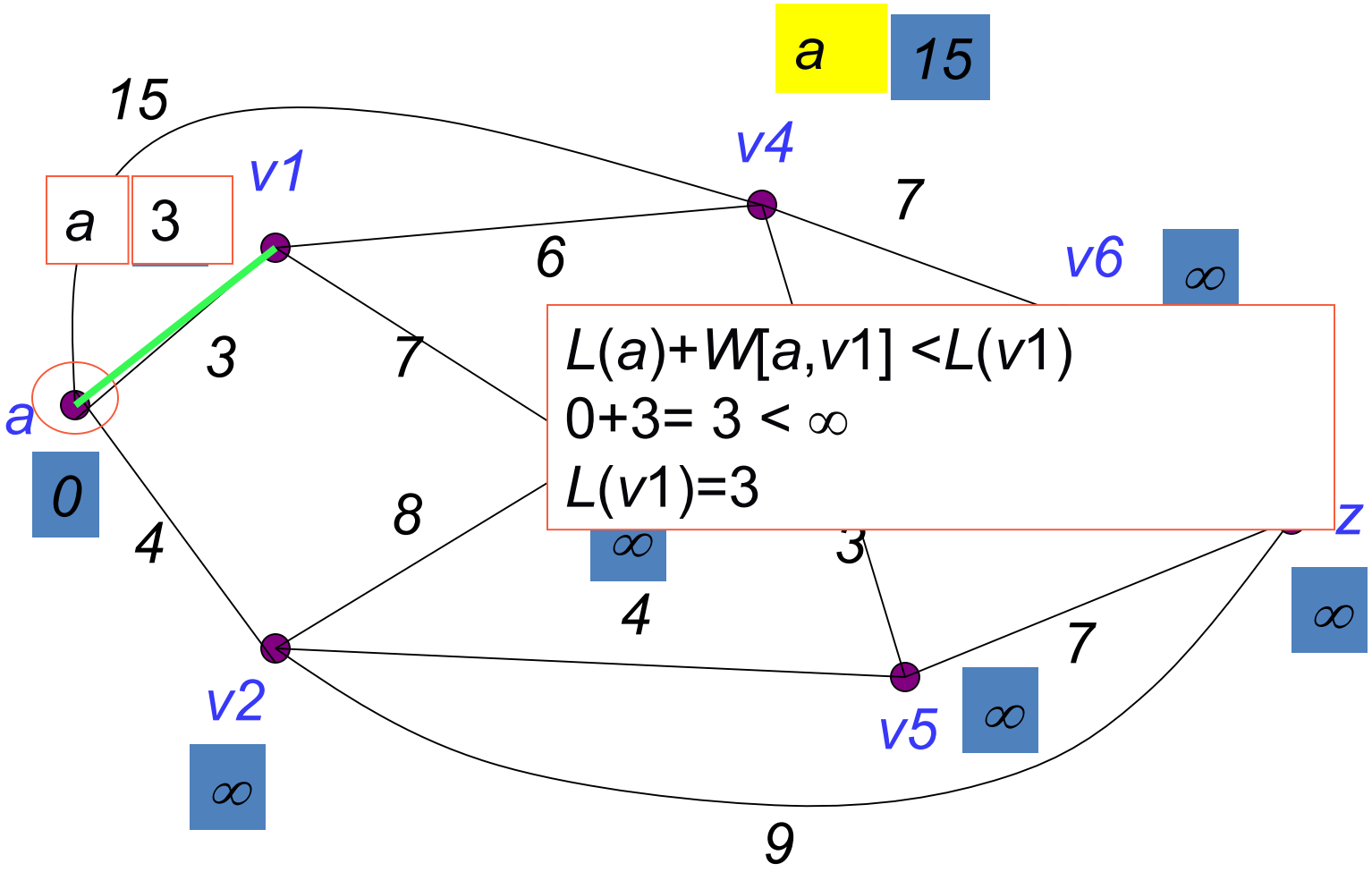
$L(w)=L(v)+W[v,w]$

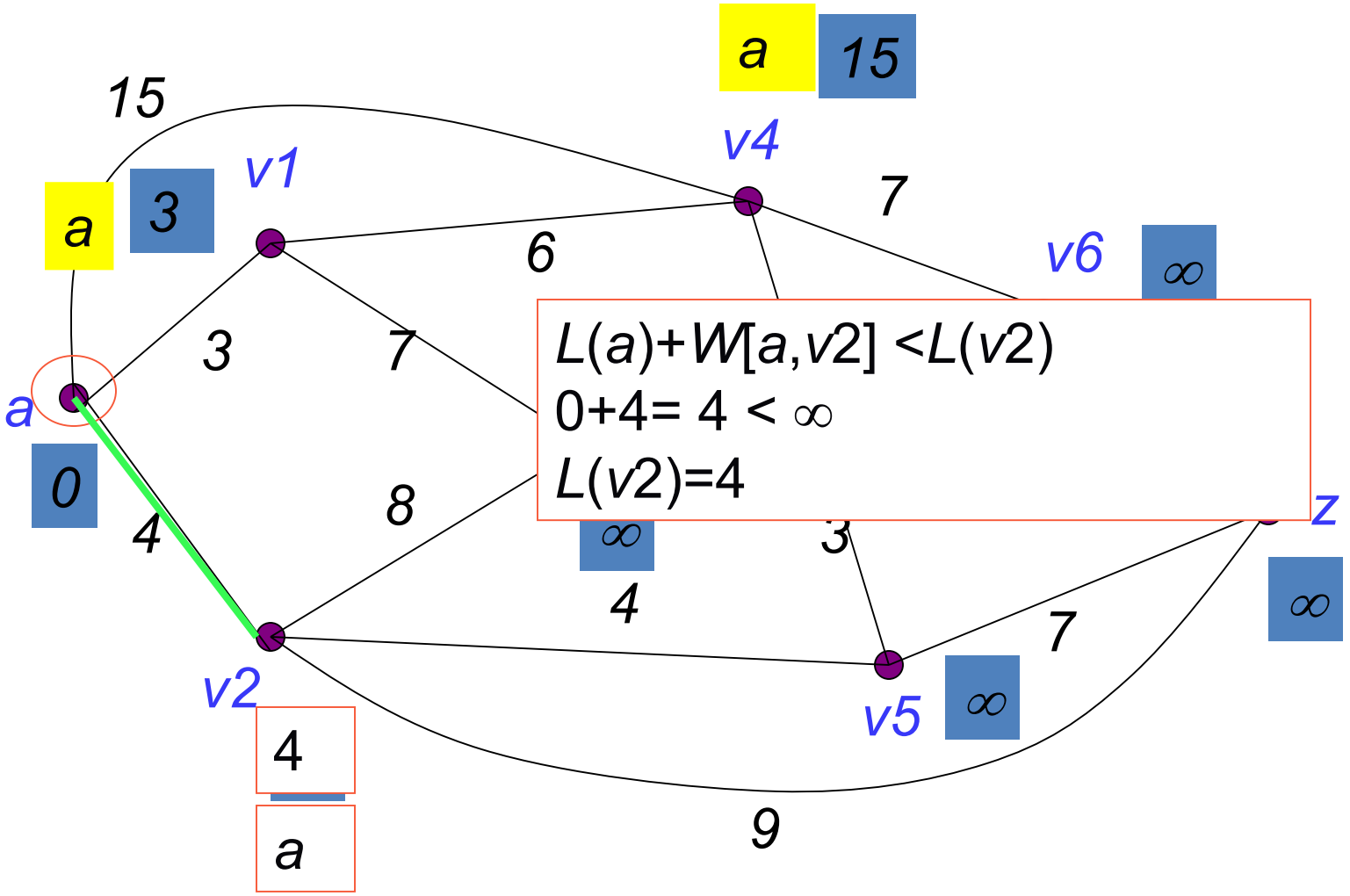
$S=\varnothing$

$N=\{a,v1,v2,v3,v4,v5,v6,z\}$




a
15
15
v4
v1
∞
6
7
v6
∞
3
7
L(a)+W[a,v4] <L(v4)
0+15=15 < ∞
L(v4)=15
a
0
8
4
∞
4
3
z
∞
7
v2
∞
v5
∞
9

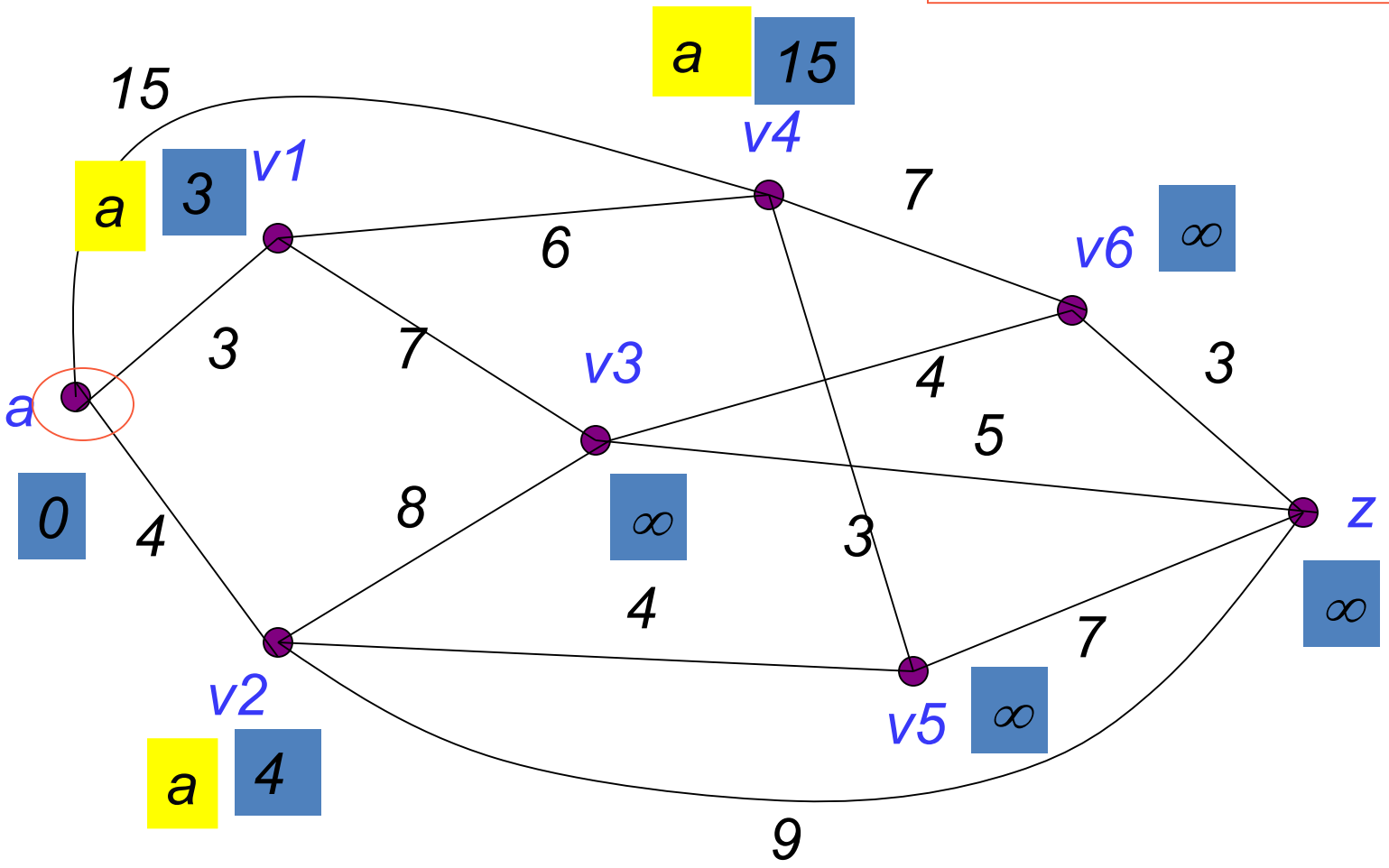

a
15
15
v4
v1
a
3
7
6
v6
∞
L(a)+W[a,v1] <L(v1)
0+3= 3 < ∞
L(v1)=3
3
7
a
0
8
4
∞
4
3
z
∞
7
v2
∞
v5
∞
9


a
15
15
v4
v1
a
3
7
6
v6
∞
L(a)+W[a,v2] <L(v2)
0+4= 4 < ∞
L(v2)=4
3
7
a
0
8
z
∞
3
4
4
7
∞
v2
v5
∞
4
a
9

$S=\{a\}$

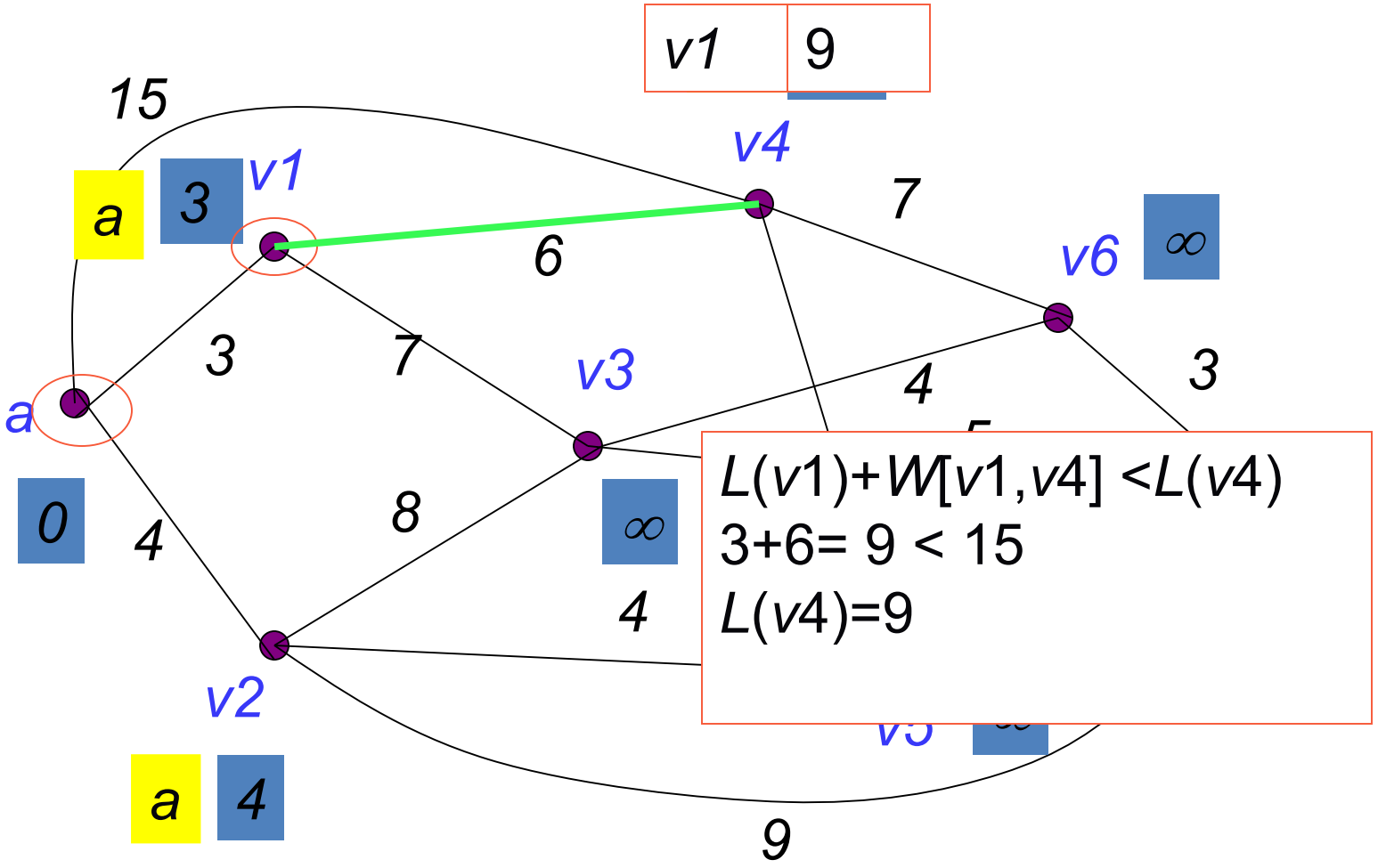
$N=\{v1,v2,v3,v4,v5$

choose $v1$
because $L(v1)=3$
$= \min\{L(u)|u \in N\}$



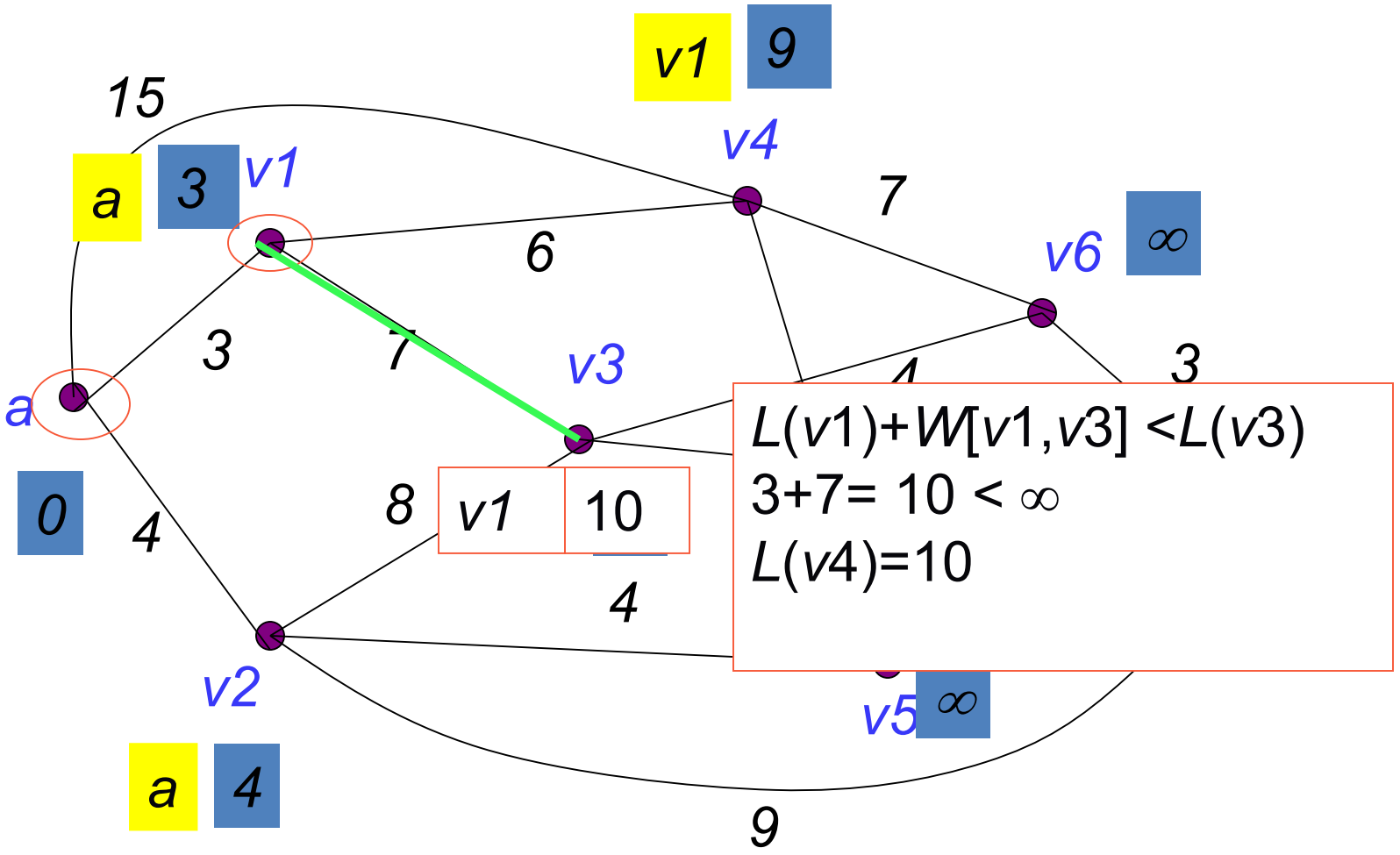
$S=\{a,v1\}$

$N=\{v2,v3,v4,v5,v6,z\}$



$S=\{a,v1\}$

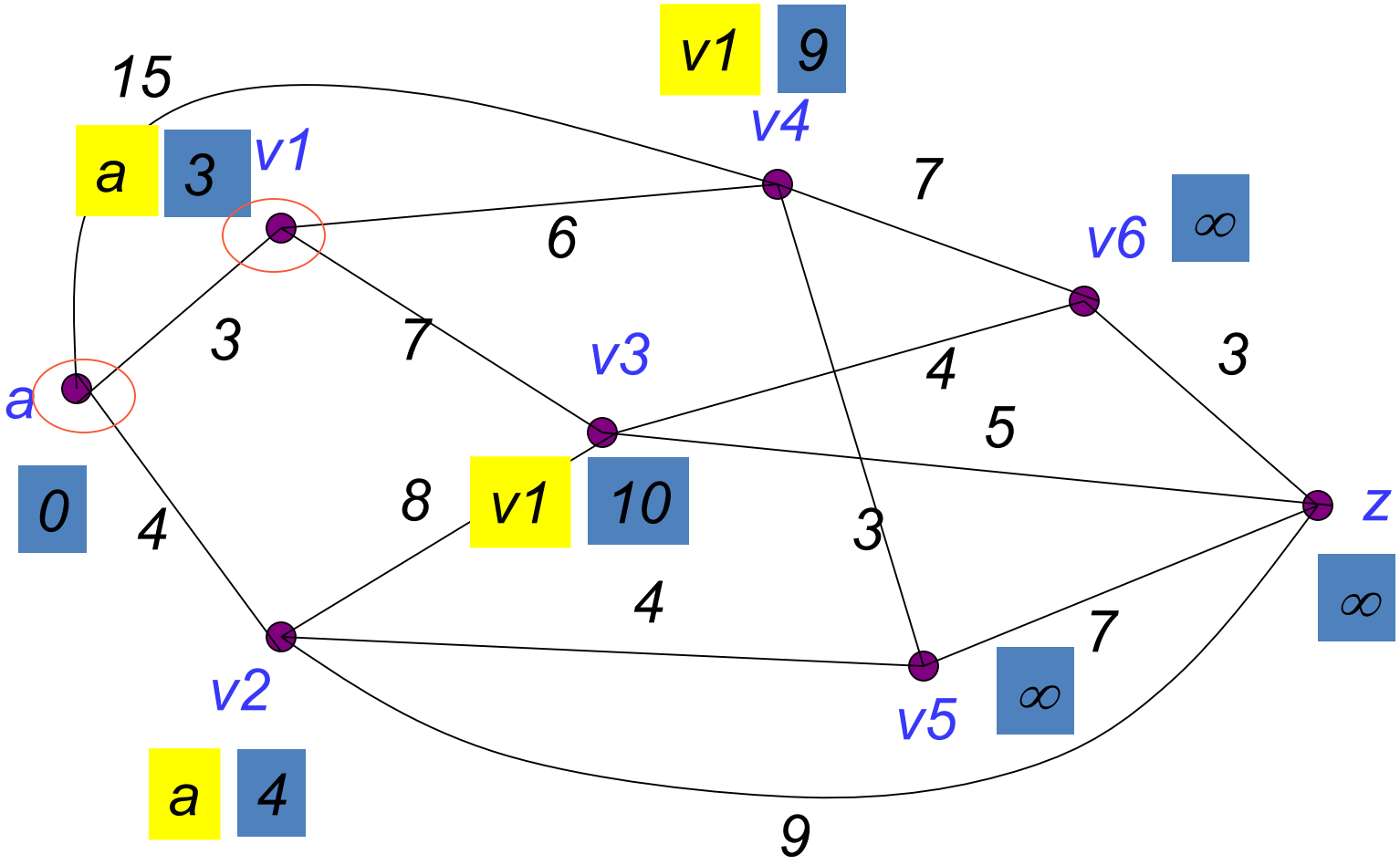
$N=\{v2,v3,v4,v5,v6,z\}$



$S=\{a, v1\}$

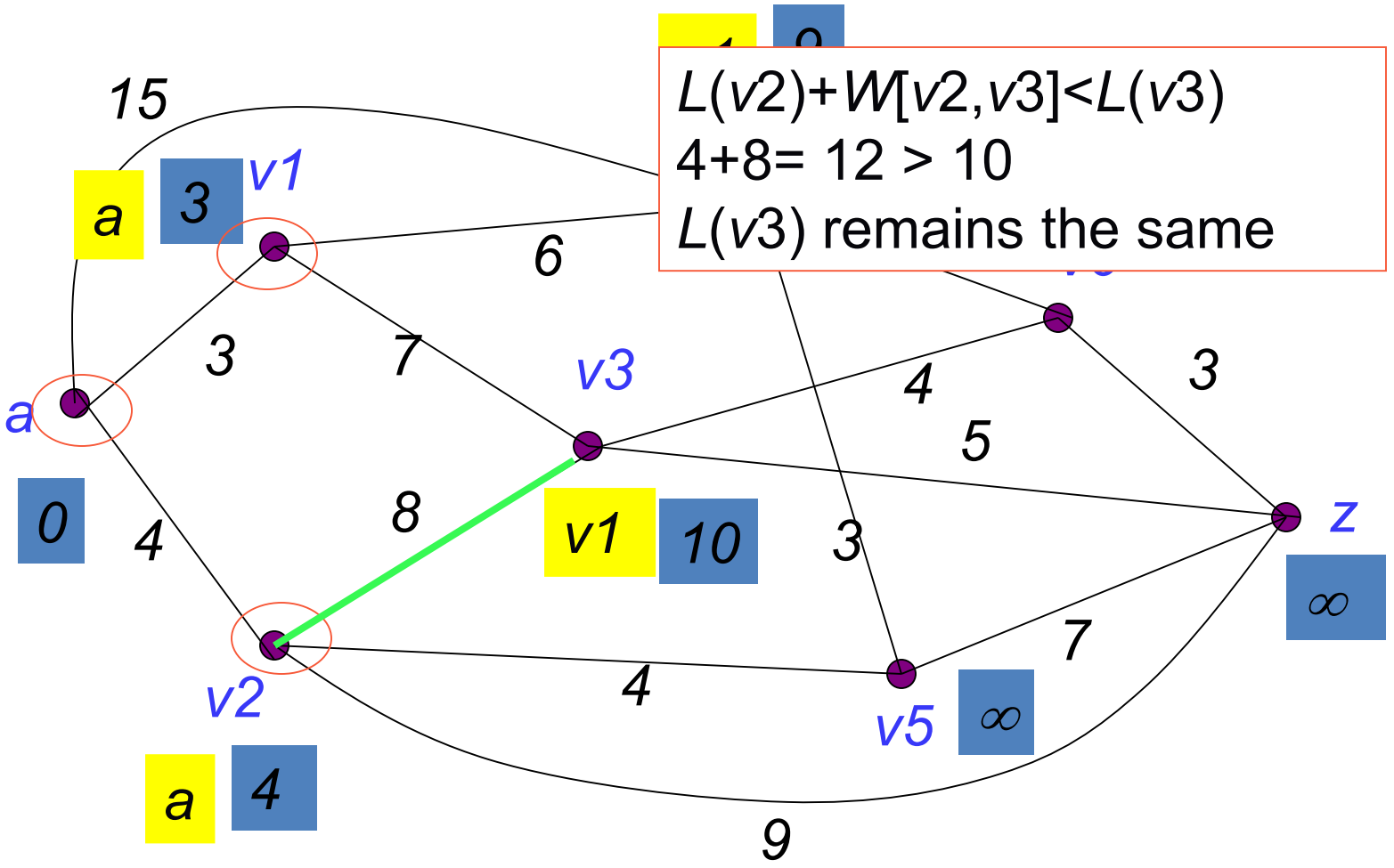
$N= \{v2,v3,v4,v5,v6,z\}$

choose $v2$
because $L(v2)=4$ =
$\min\{L(u)|u\in N\}$



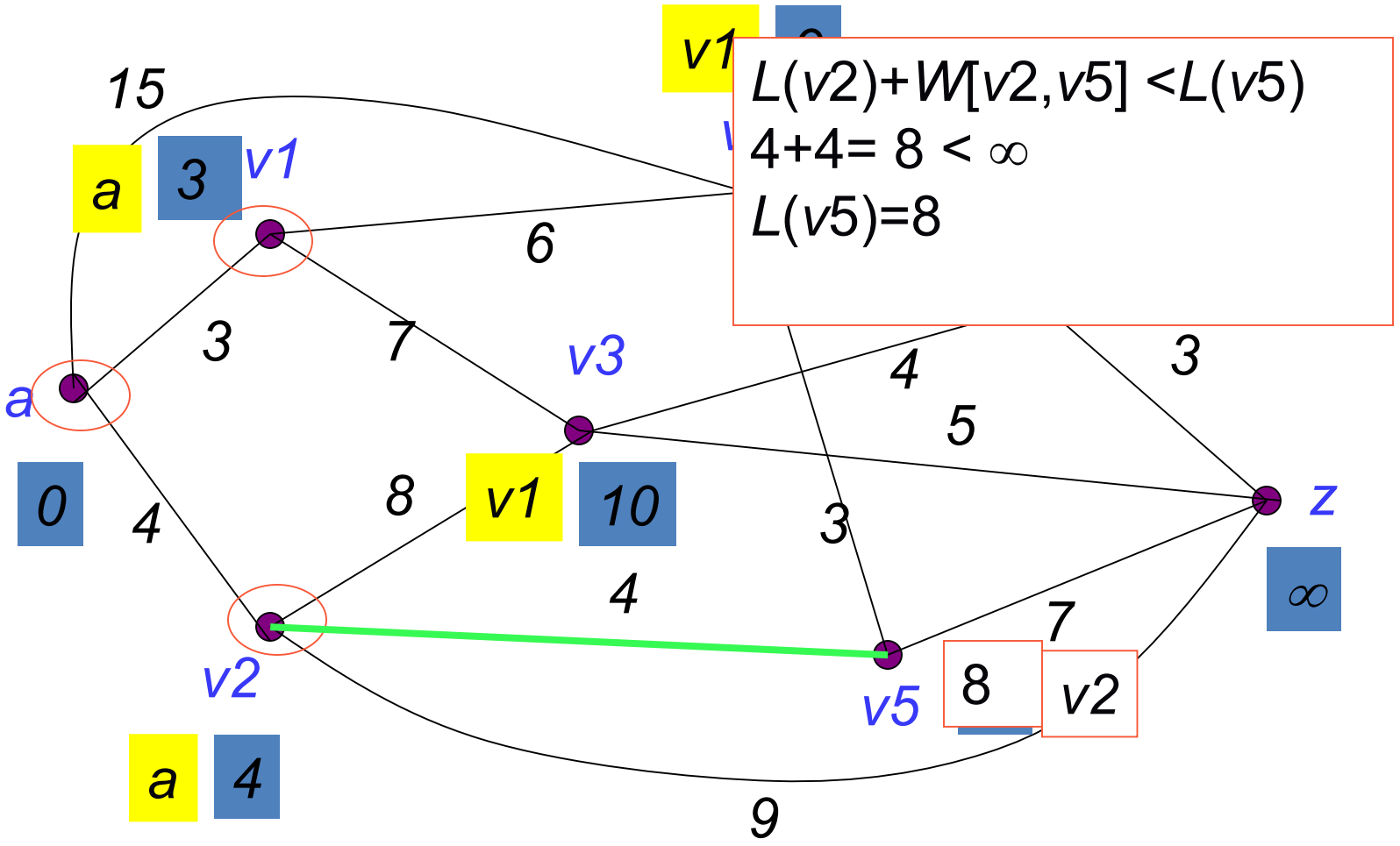
$S=\{a, v1, v2\}$

$N=\{v3,v4,v5,v6,z\}$



$S=\{a, v1, v2\}$

$N= \{v3,v4,v5,v6,z\}$



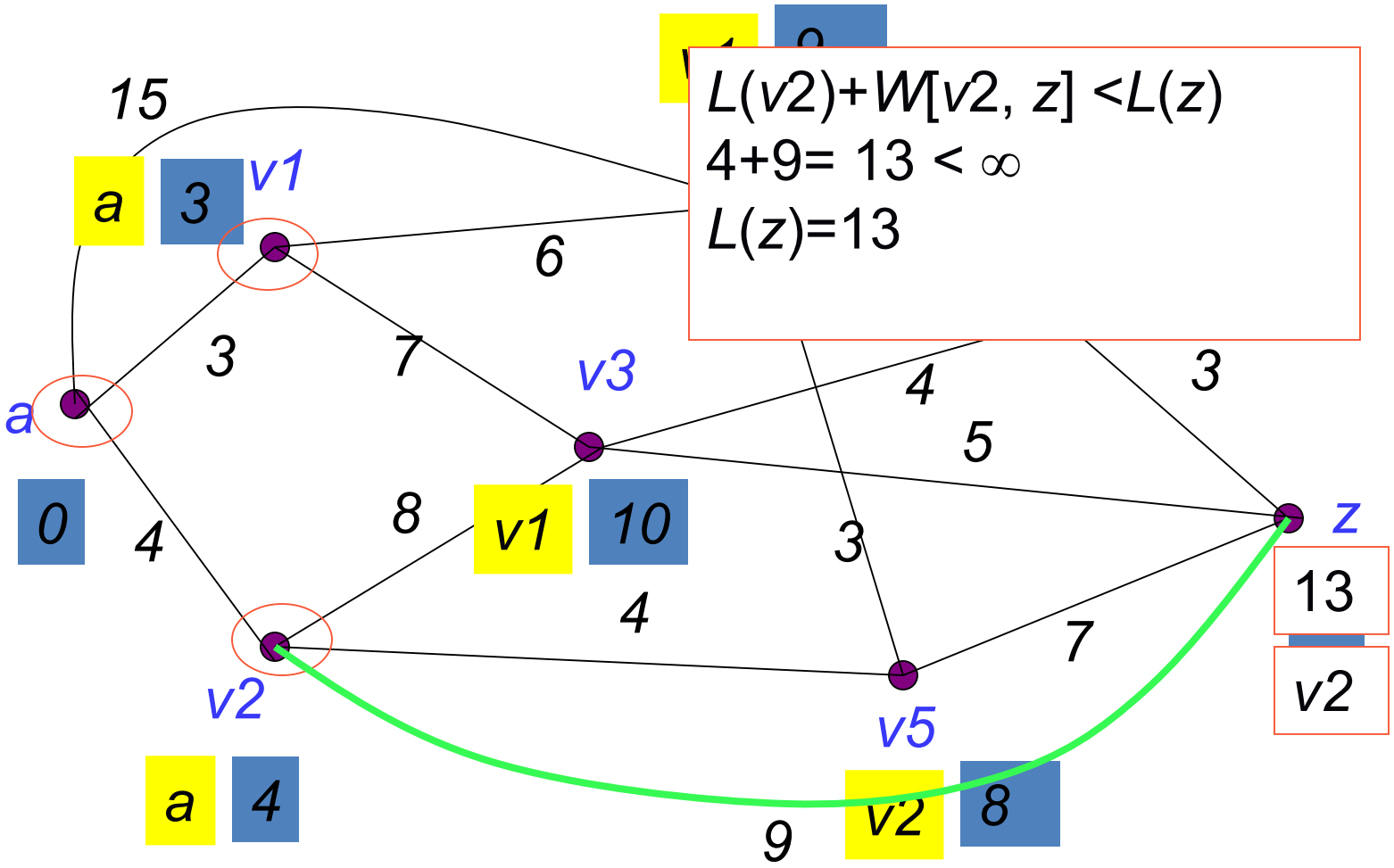
$L(v2)+W[v2,v5] < L(v5)$

$4+4= 8 < \infty$

$L(v5)=8$

$S=\{a, v1, v2\}$

$N= \{v3,v4,v5,v6,z\}$



$L(v2)+W[v2, z] < L(z)$

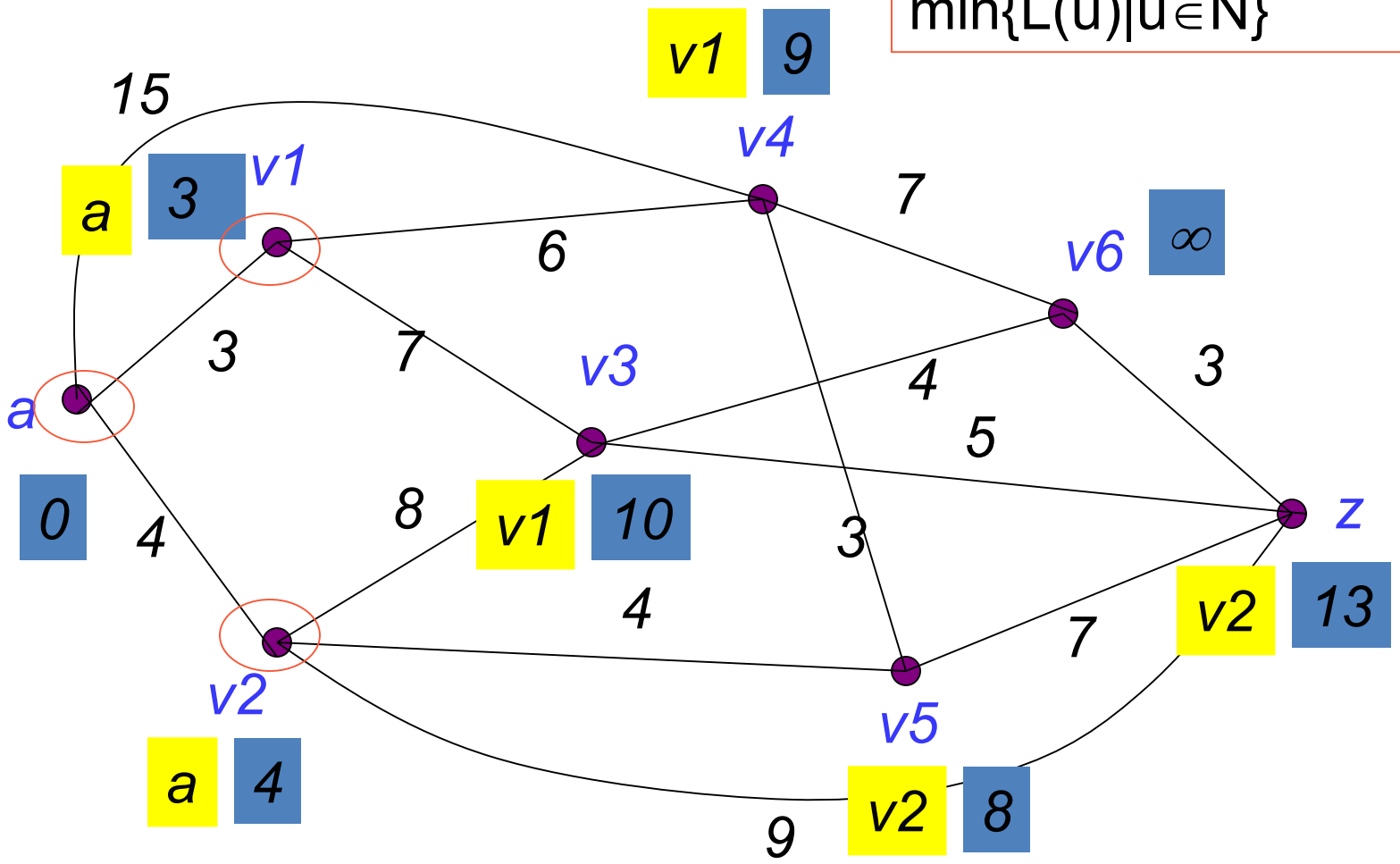
$4+9= 13 < \infty$

$L(z)=13$

$S=\{a,v1,v2\}$

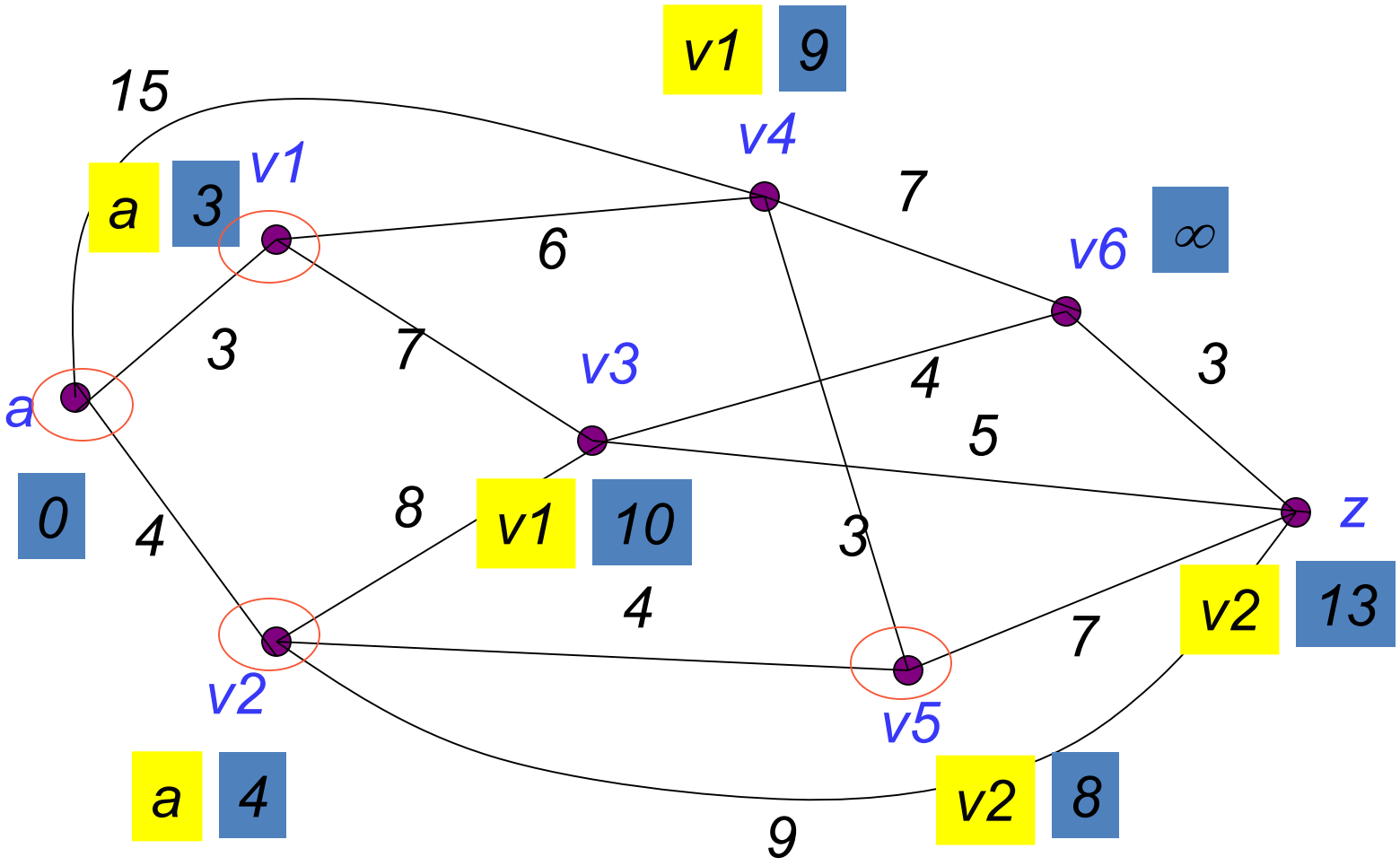
$N=\{v3,v4,v5,v6,z\}$

choose $v5$ because $L(v5)=8 = \min\{L(u)|u\in N\}$



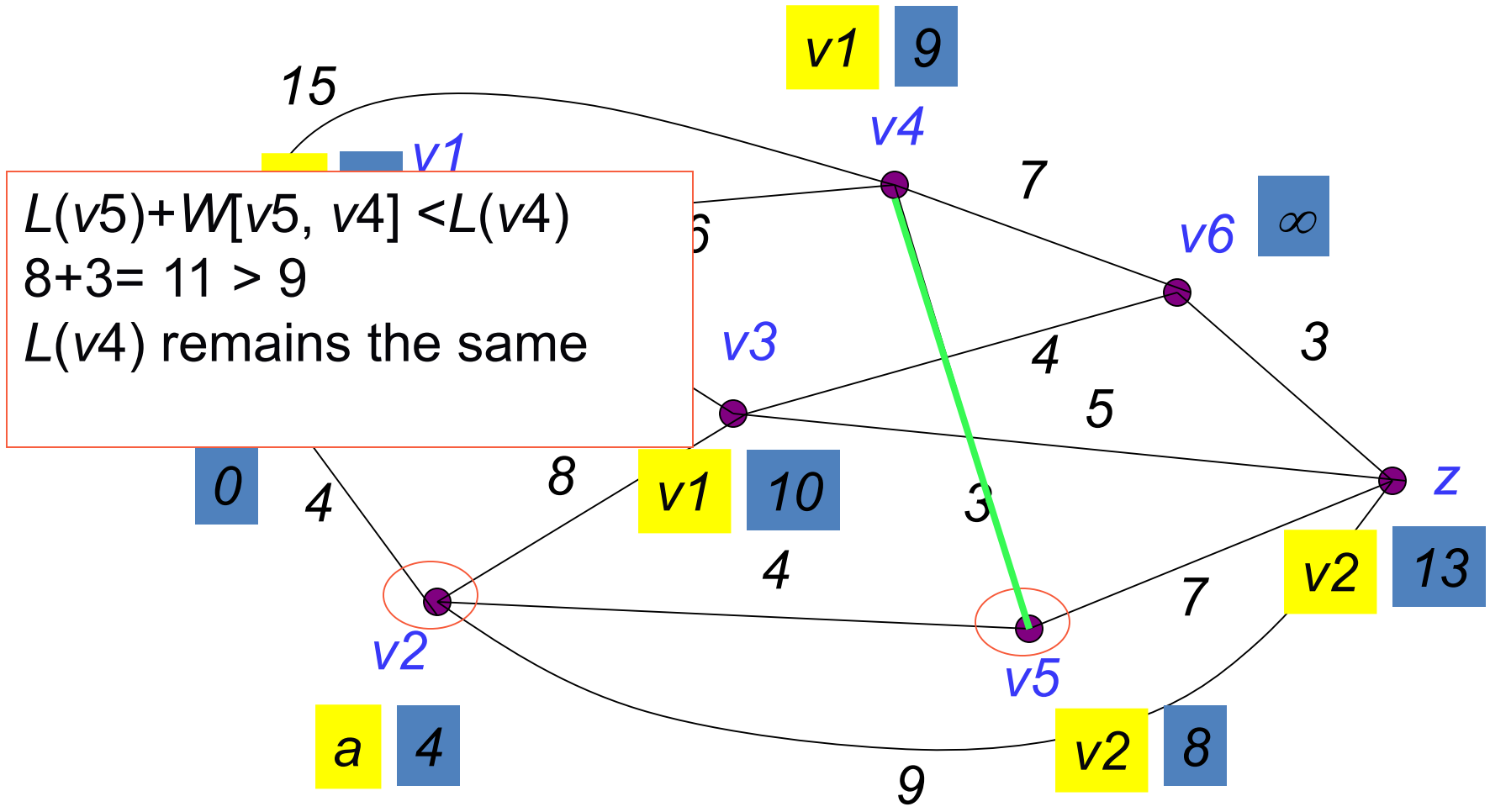
S={a,$v1$,$v2$,$v5$}

N= {$v3$,$v4$,$v6$,z}



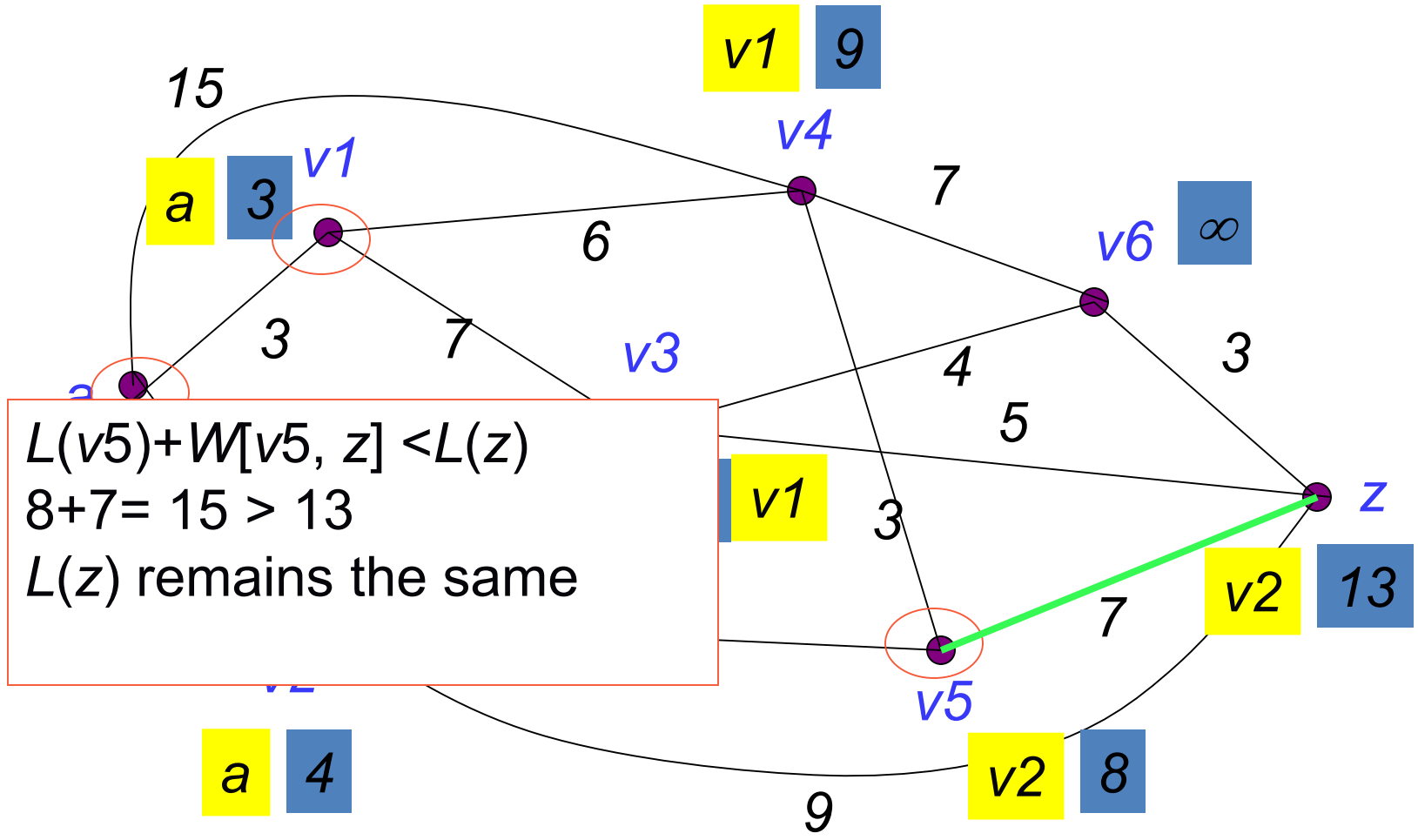
$S=\{a,v1,v2,v5\}$

$N=\{v3,v4,v6,z\}$



$S=\{a,v1,v2,v5\}$

$N=\{v3,v4,v6,z\}$



$L(v5)+W[v5, z] < L(z)$

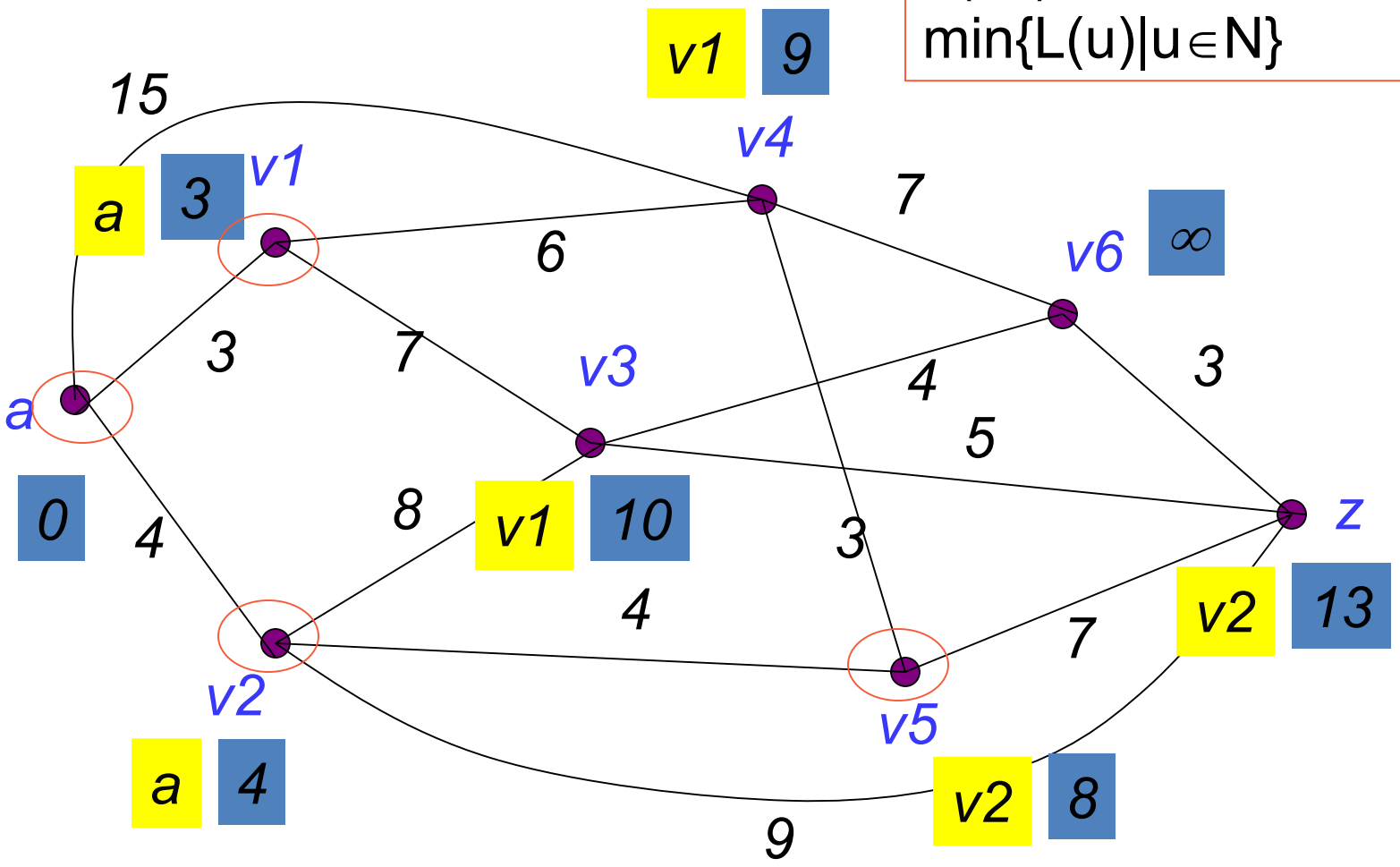
$8+7= 15 > 13$

$L(z)$ remains the same

S={a,$v1$,$v2$,$v5$}

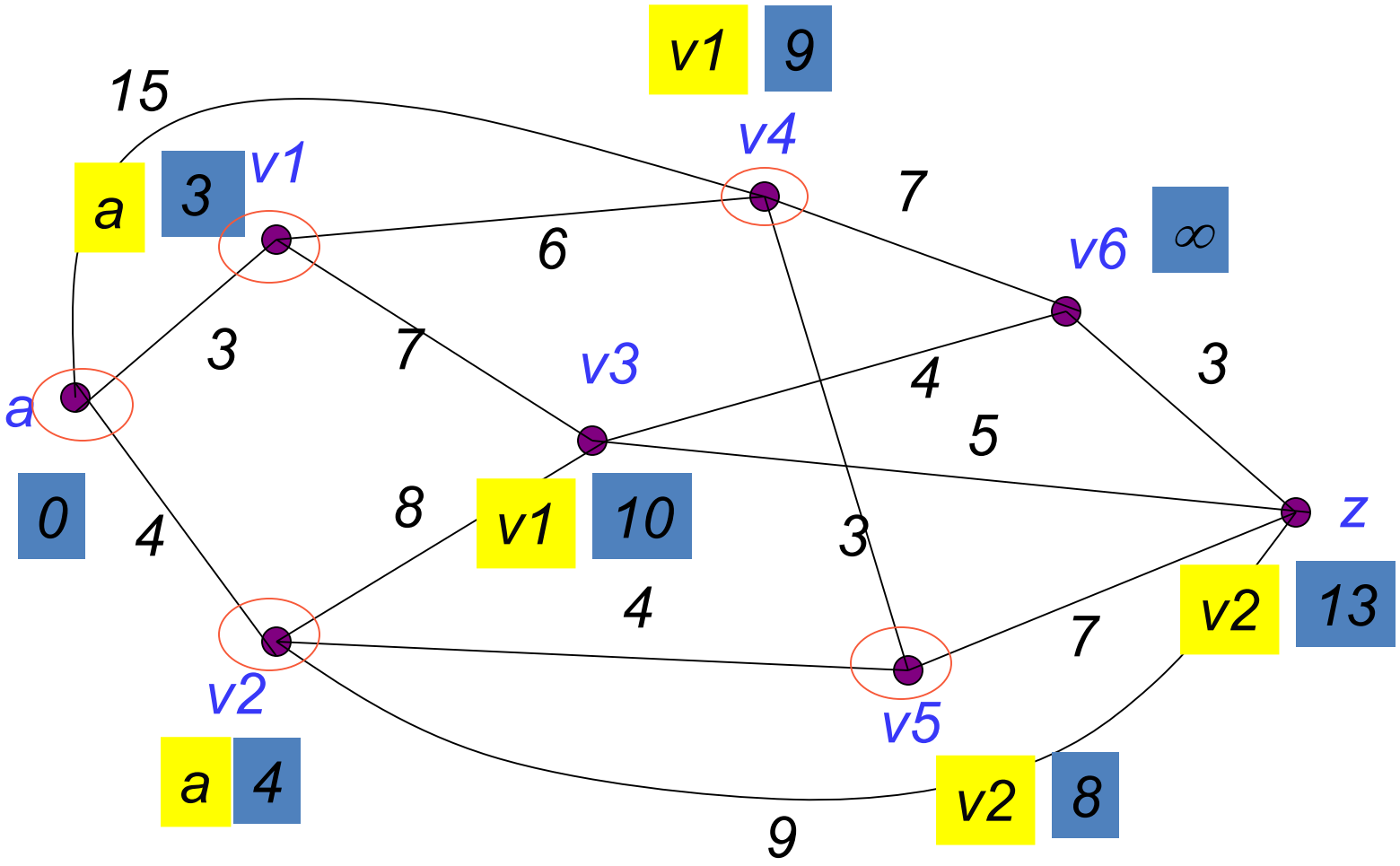
N= {$v3$,$v4$,$v6$,z}

choose $v4$ because $L(v4)=9$ = min{L(u)|u∈N}



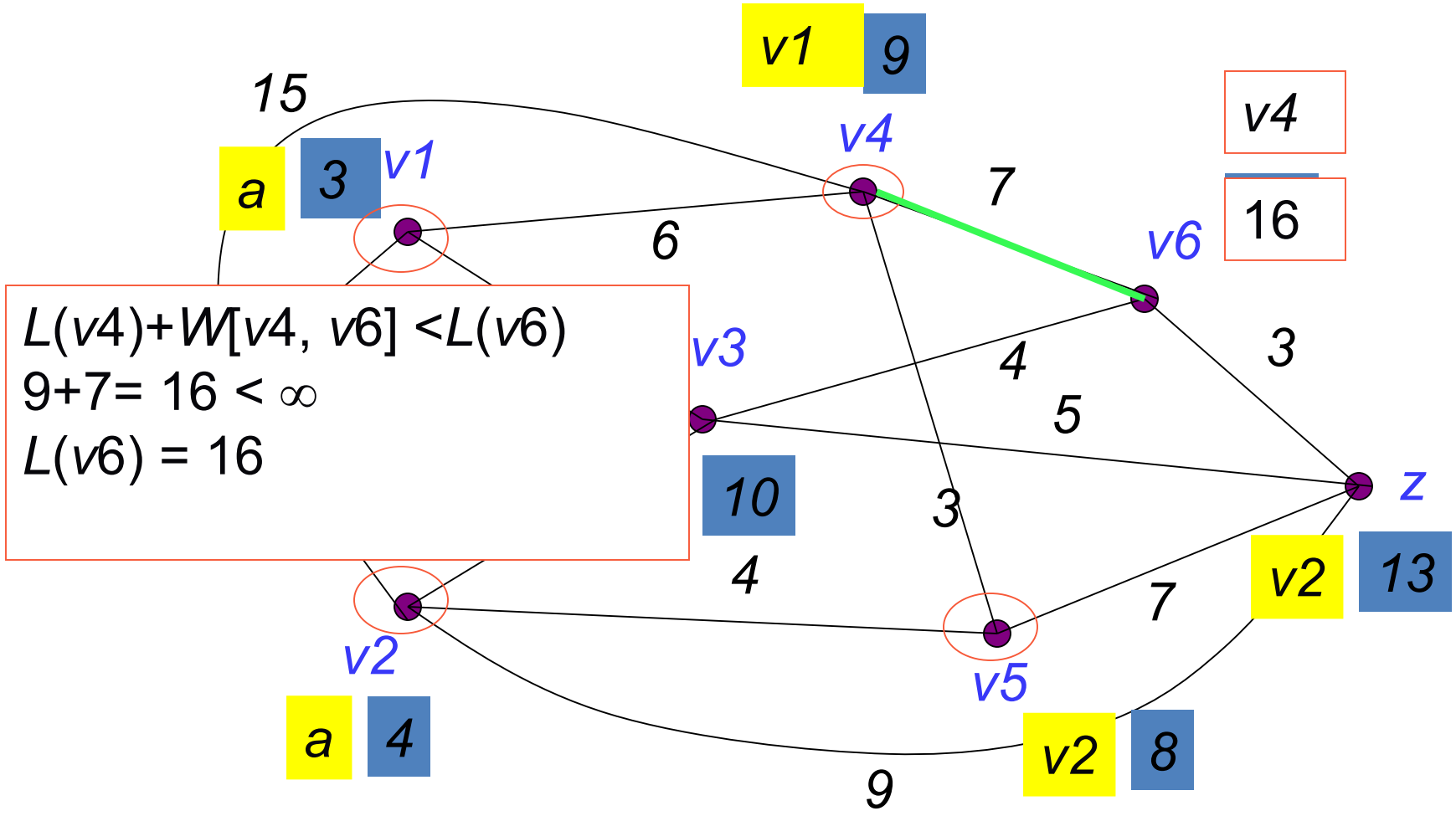
S={a,$v1$,$v2$, $v4$, $v5$}

N= {$v3$, $v6$, z}



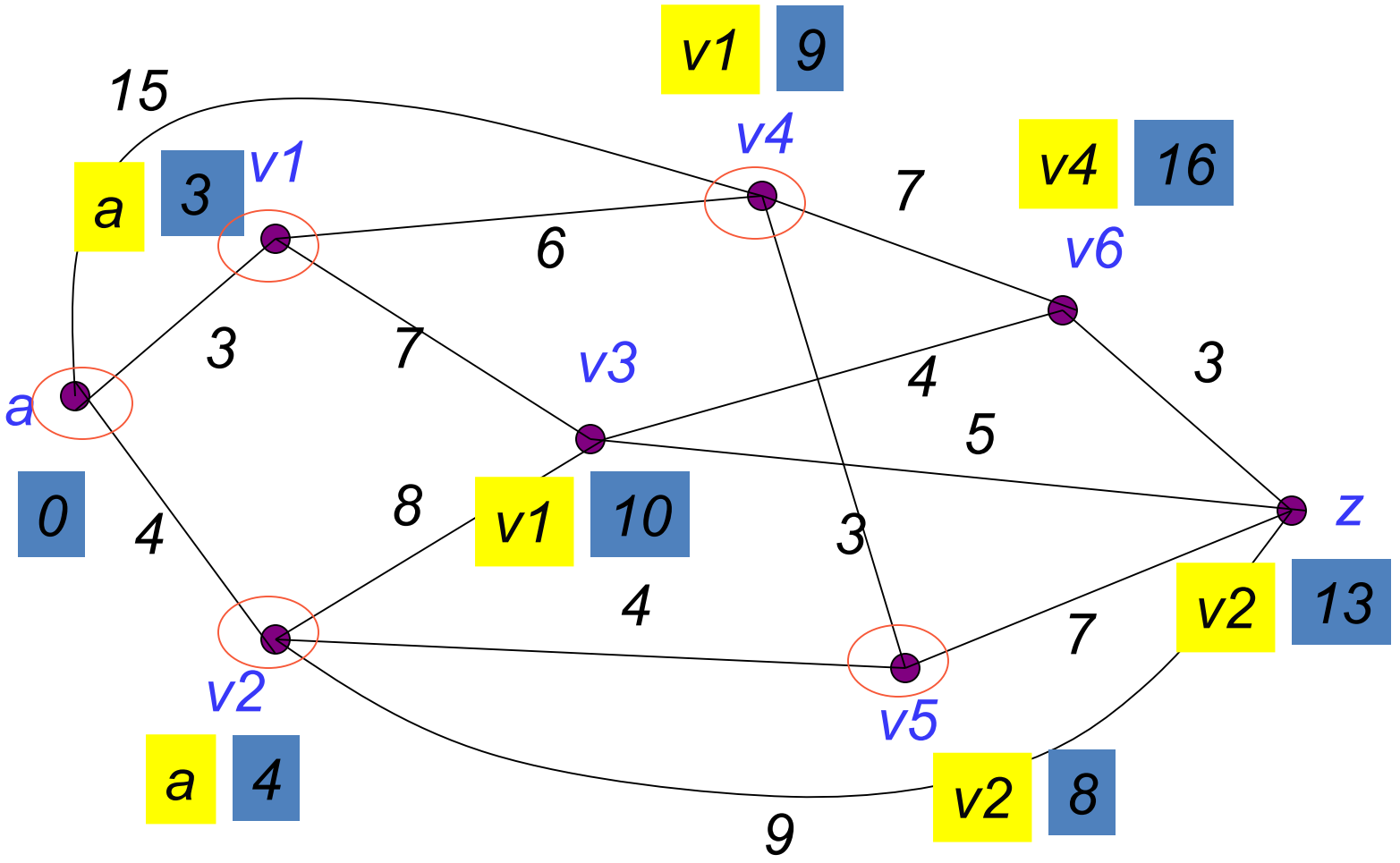
$S=\{a, v1, v2, v4, v5\}$

$N=\{v3, v6, z\}$



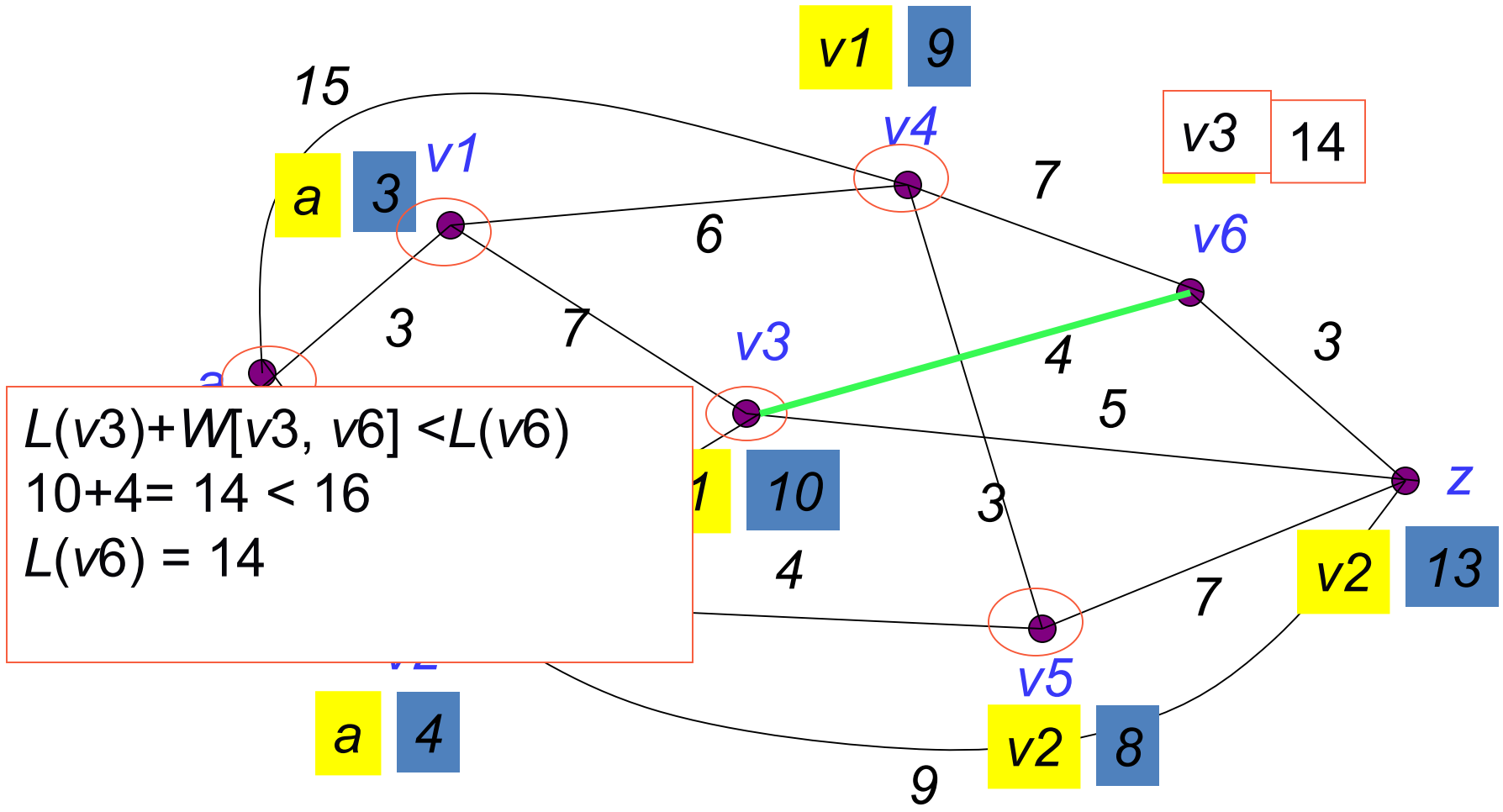
$S=\{a, v1, v2, v4, v5\}$
$N=\{v3, v6, z\}$

choose $v3$
because $L(v3)=10$
$= \min\{L(u) | u \in N\}$



$S=\{a, v1, v2, v3, v4, v5\}$

$N=\{v6, z\}$



$L(v3)+W[v3, v6] < L(v6)$

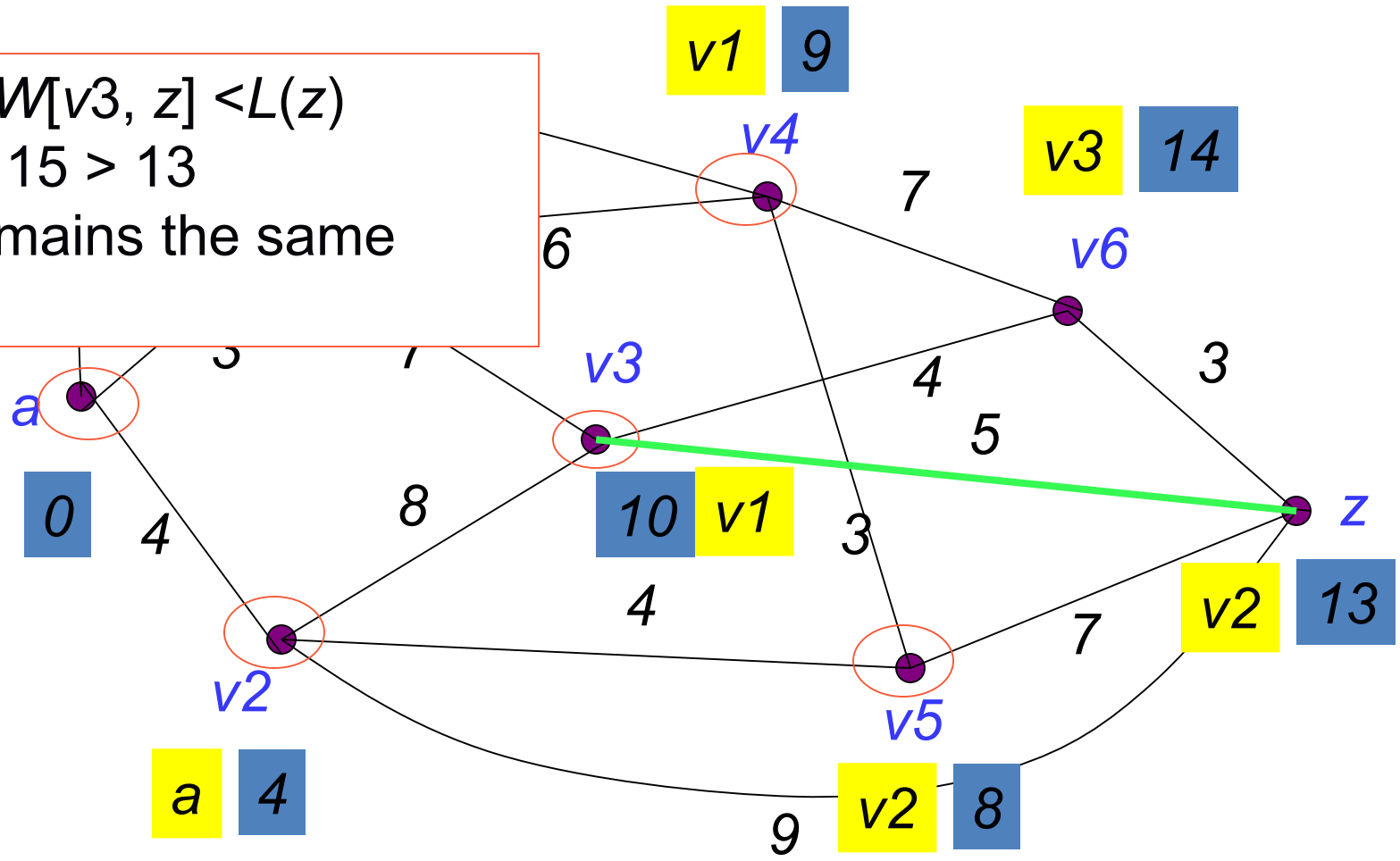
$10+4 = 14 < 16$

$L(v6) = 14$

S={a,$v1$,$v2$, $v3$, $v4$,$v5$}

N= {$v6$,z}

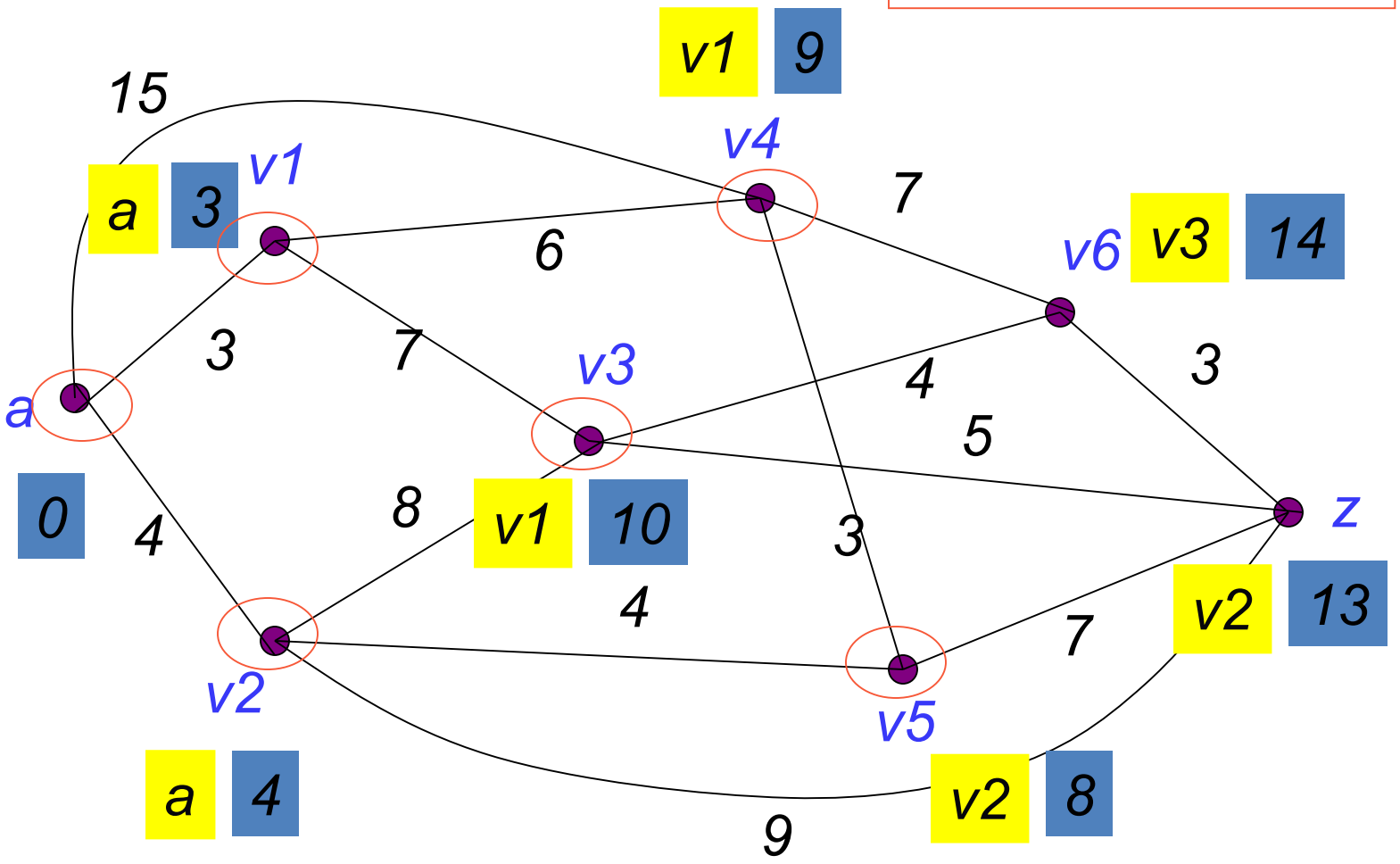
$L(v3)+W[v3, z] < L(z)$
10+5= 15 > 13
$L(z)$ remains the same



$S=\{a,v1,v2,v3,v4,v5\}$

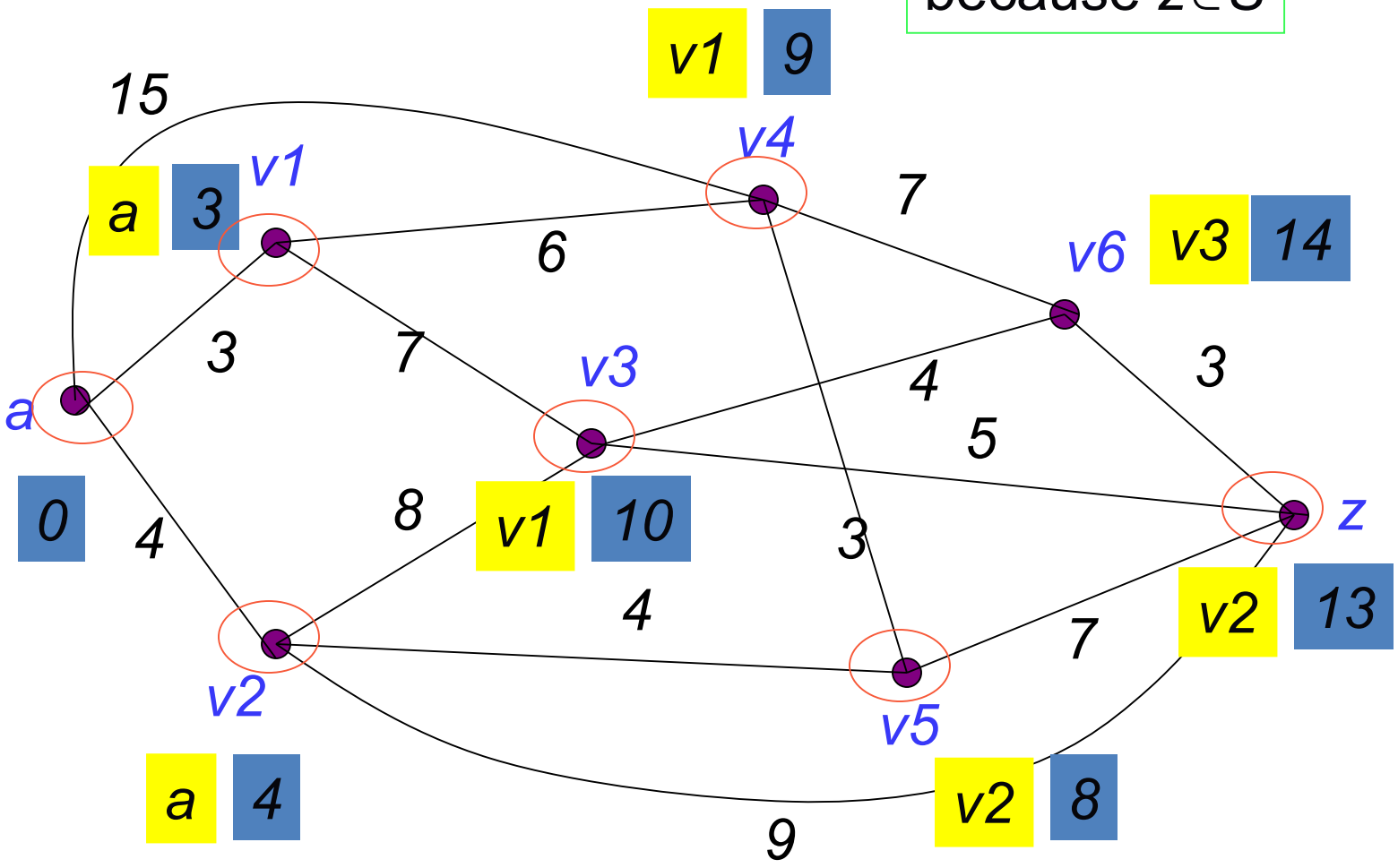
$N=\{v6,z\}$

choose z because
$L(z)=13 =$
$\min\{L(u)|u\in N\}$



S={a,$v1$,$v2$,$v3$,$v4$,$v5$, z}
N= {$v6$}

The loop terminates because $z \in S$



Shortest path from *a* to *z*

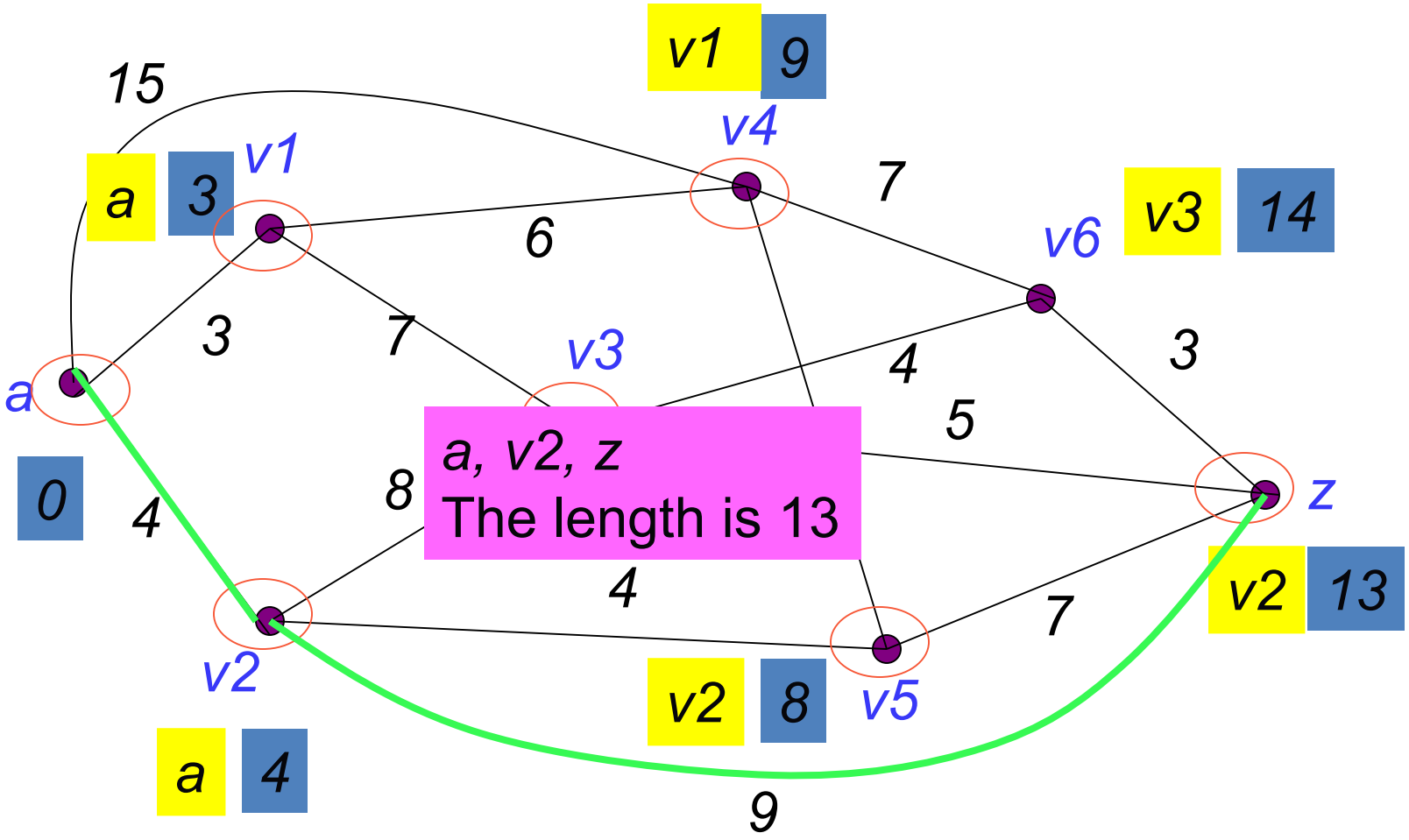
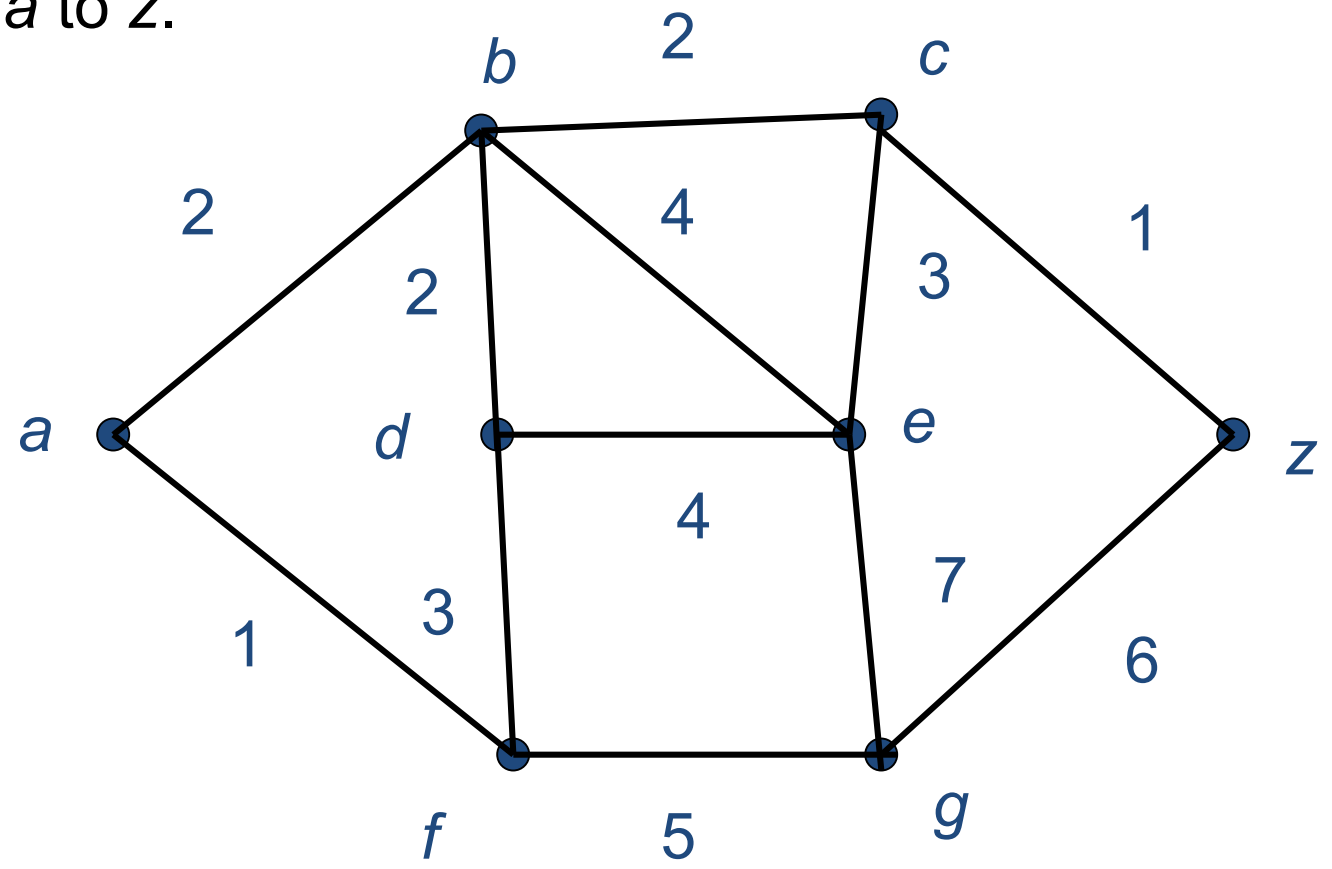


Table – Djikstra Algorithm

No.	S	N	L(a)	$L(V_1)$	$L(V_2)$	$L(V_3)$	$L(V_4)$	$L(V_5)$	$L(V_6)$	L(z)
0	{ }	{a, V_1, V_2, V_3, V_4, V_5,V_6, z }	0	∞	∞	∞	∞	∞	∞	∞
1	{a}	{V_1, V_2, V_3, V_4, V_5,V_6, z }		3	4	∞	15	∞	∞	∞
2	{a, V_1}	{V_2, V_3, V_4, V_5,V_6, z }		**3**	4	10	9	∞	∞	∞
3	{a, V_1, V_2}	{V_3, V_4, V_5,V_6, z }			**4**	10	9	8	∞	13
4	{a, V_1, V_2, V_5}	{V_3, V_4,V_6, z }				10	9	**8**	∞	13
5	{a, V_1, V_2, V_5, V_4}	{V_6, z }				10	**9**		16	13
6	{a, V_1, V_2, V_5, V_4, V_3,}	{V_6, z }				**10**			14	13
7	{a, V_1, V_2, V_5, V_4, V_3, z}	{V_6 }							14	**13**

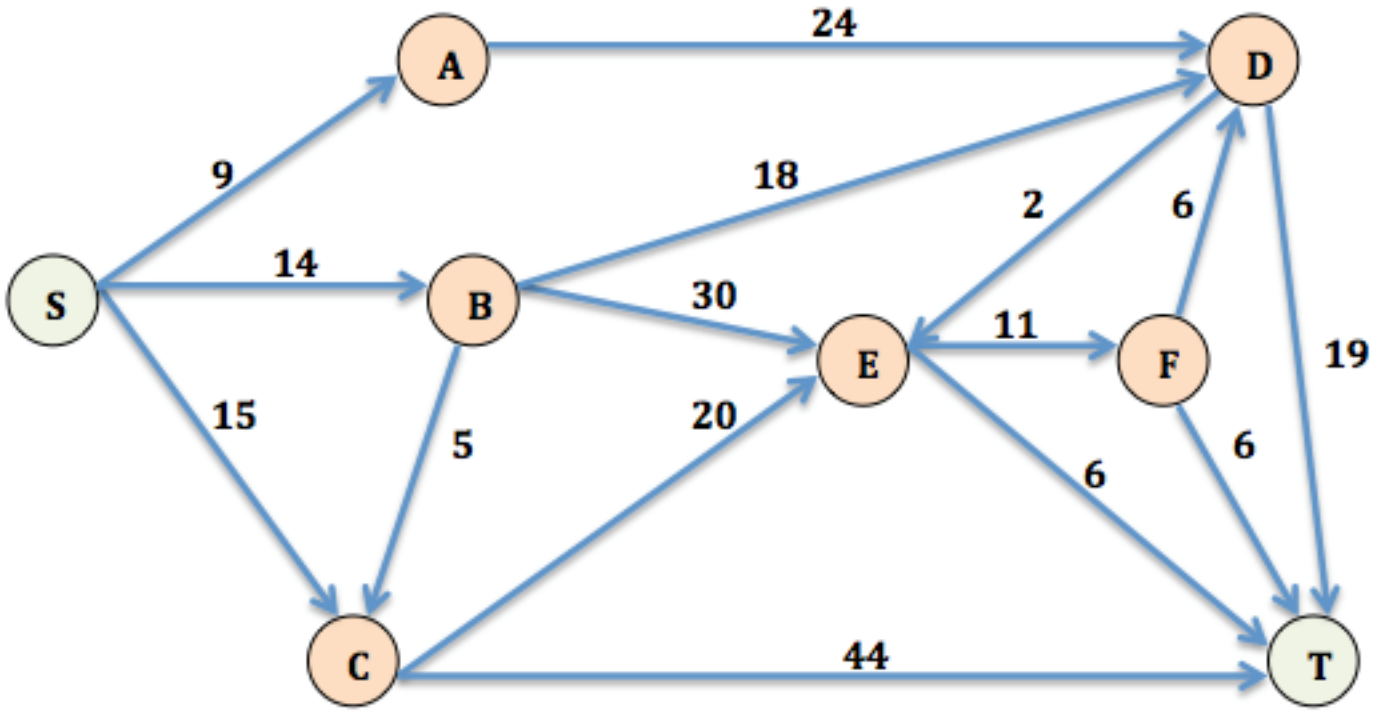
exercise

Use Dijkstra's algorithm to find the length of a shortest path from *a* to *z*.



exercise

Q: Given a weighted digraph, find the shortest path from S to T, using Djik



Note: Weights are arbitrary numbers (i.e., not necessarily distances).

Exercise Past Year 2015/2016

The network in Figure 5 gives the distances in miles between pairs of cities A, B, …, and H.

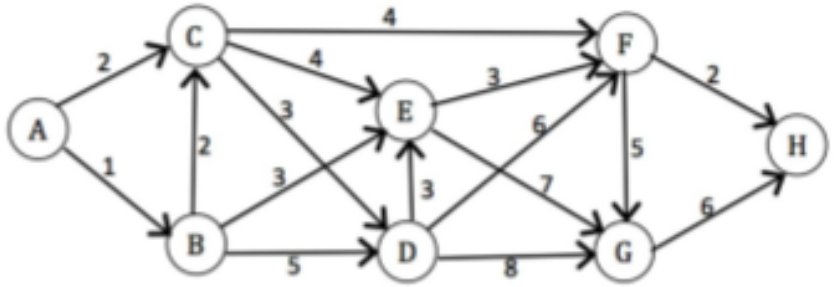


Figure 5

a) Based on Dijkstra's algorithm, complete Table 1 to find the shortest path from city A to city H. (Note: Copy Table 1 into your answer booklet).

(8 marks)

Table 1

Iteration	S	N	L(A)	L(B)	L(C)	L(D)	L(E)	L(F)	L(G)	L(H)
0										
1										
2										
3										
4										
5										
6										
7										

b) State the minimum distance and the shortest path from city 1 to city 8.

(2 marks)